

I collaborated with Giselle and Allison in completing this problem set!

1 Final Project

Giselle and I are interested in working together on a project related to planetary auroral radio emissions. The following is taken from what she wrote since she has a much more comprehensive understanding of the astrophysical idea, but I of course plan to get up-to-speed on the astrophysics and contribute with my CS knowledge (in e.g. building efficient simulators, working with software packages, and processing data).

For the final project, I am interested in planetary auroral radio emission and what it lets us infer about a planet's magnetic field. Radio emission produced through the cyclotron maser instability (CMI) is helpful because its frequency is directly related to the local magnetic-field strength where the radio emission is generated. However, what an observer actually sees also depends on which part of the magnetosphere is producing the emission and also the viewing geometry, being that CMI emission is strongly beamed. Jupiter is a good system for exploring this because it has both auroral radio emission associated with its general magnetospheric system and radio emission specifically connected to its interaction with Io[4]. Both of these two types of emission occur within the same underlying magnetic field but they sample different magnetic field lines. A current research question may be: how much magnetic-field information could a distant, unresolved observer recover from Jupiter's radio spectrum, and how does it change between Io-controlled and main-auroral emission or with viewing angle? I am especially interested in this since planetary magnetic fields help us learn about planetary interiors and magnetospheres, and radio observations might eventually be one of the few ways to constrain magnetic fields for exoplanets that cannot be spatially resolved.

A main tool I am considering is ExPRES (Exoplanetary and Planetary Radio Emissions Simulator), which models CMI radio emission based on the planetary magnetic field, the source location, electron properties, beaming geometry, and also the observer's location[3]. For Jupiter's magnetic field, we could use JRM33, which is a modern field model derived from Juno magnetic-field measurements[1]. JRM33 would be useful for this since Jupiter's real magnetic field is much more complicated than a simple tilted dipole, so we can treat it as our "ground truth" and then ask what a distant observer would recover from radio measurements. For the Io-controlled case, recent Juno observations have directly characterized radio source regions associated with Io, Europa, and Ganymede and also confirmed loss-cone CMI as the radio-generation mechanism[2]. For the more general Jovian auroral case, earlier ExPRES studies have modeled sources associated with Jupiter's main auroral region, so there is already a physically motivated starting configuration. We could begin by working mainly with simulations and published constraints, and then use Juno/Waves or other Jovian radio observations for validation if it seems feasible in the time frame.

The biggest challenge is that planetary radio emission depends on a lot of parameters at once, including the magnetic-field geometry, source field line, electron energy, plasma

conditions, beaming angle, and also where the observer is located. This means we do not want to try and build a full Jupiter magnetosphere model but rather the idea would be to hold most of the physics constant and run a controlled comparison. We could use the same JRM33 magnetic field for every simulation and initially compare only an Io-attached source and a main-auroral source, while testing a small set of observer inclinations. Something else we will need to think about is what recovering the magnetic field will look like. The radio frequency directly gives the local field strength at the emitting source, but it is not automatically the same as Jupiter’s global dipole strength or maximum surface field. Our project would probably focus on what magnetic-field constraints remain reliable from an unresolved spectrum, rather than trying to retrieve one single value for Jupiter’s entire field. We would also keep other moons, plasma variability, and raw Juno data analysis as potential extensions.

References

- [1] J. E. P. Connerney, S. Timmins, R. J. Oliverson, J. R. Espley, J. L. Joergensen, S. Kotsiaros, P. S. Joergensen, J. M. G. Merayo, M. Herceg, J. Bloxham, K. M. Moore, A. Mura, A. Moirano, S. J. Bolton, and S. M. Levin. A new model of jupiter’s magnetic field at the completion of juno’s prime mission. *Journal of Geophysical Research: Planets*, 127(2):e2021JE007055, 2022. e2021JE007055 2021JE007055.
- [2] C. K. Louis, P. Louarn, B. Collet, N. Clément, S. Al Saati, J. R. Szalay, V. Hue, L. Lamy, S. Kotsiaros, W. S. Kurth, C. M. Jackman, Y. Wang, M. Blanc, F. Allegrini, J. E. P. Connerney, and D. Gershman. Source of radio emissions induced by the galilean moons io, europa and ganymede: In situ measurements by juno. *Journal of Geophysical Research: Space Physics*, 128(12):e2023JA031985, 2023. e2023JA031985 2023JA031985.
- [3] Corentin Louis, Sebastien Hess, Baptiste Cecconi, Philippe Zarka, Laurent Lamy, S. Aicardi, and A. Loh. Expres: an exoplanetary and planetary radio emissions simulator. *Astronomy Astrophysics*, 627:A30, 06 2019.
- [4] P. Zarka, F. P. Magalhães, M. S. Marques, C. K. Louis, E. Echer, L. Lamy, B. Cecconi, and R. Prangé. Jupiter’s auroral radio emissions observed by cassini: Rotational versus solar wind control, and components identification. *Journal of Geophysical Research: Space Physics*, 126(10):e2021JA029780, 2021. e2021JA029780 2021JA029780.

2 Derive Kepler’s Equation

Starting with Kepler’s Second Law, we know that $\frac{dA}{dt}$ is a constant. It therefore must be equal to the total area of the ellipse divided by the period:

$$\frac{dA}{dt} = \frac{\pi ab}{P}. \quad (1)$$

Substituting into this the mean motion $n = \frac{2\pi}{P}$ and noting that $\frac{dM}{dt} = n$, we can get an expression relating A to the mean anomaly M , which will prove quite useful in a bit:

$$\frac{dA}{dt} = \frac{1}{2}ab \frac{dM}{dt}. \quad (2)$$

In polar coordinates, a differential area is defined as the following for radius r and angle f :

$$dA = \frac{1}{2}r^2 df. \quad (3)$$

We can multiply both sides of (2) by dt and equate to (3), giving us an excitingly useful-looking expression for dM (which we will integrate later once we have nice values to substitute in!):

$$\begin{aligned} dA &= \frac{1}{2}r^2 df = \frac{1}{2}ab dM \\ \implies dM &= \frac{r^2}{ab} df. \end{aligned} \quad (4)$$

In the end, we want M as a function of E , so it would help to have a dE somewhere on the right side. We can use the chain rule:

$$dM = \frac{r^2}{ab} \frac{df}{dE} dE. \quad (5)$$

Our goal is then to find appropriate values to substitute in for our unknowns: r and $\frac{df}{dE}$. We can begin with r . We define x and y as the planet's Cartesian coordinates along the semi-major and semi-minor axes relative to the center of the ellipse. The geometry of the problem then tells us the following:

$$\cos E = \frac{x}{a}; \quad (6)$$

$$\sin E = \frac{y}{b} = \frac{y}{a\sqrt{1-e^2}}; \quad (7)$$

$$\cos f = \frac{x-c}{r} = \frac{x-ae}{r}. \quad (8)$$

We can also see that $r = \sqrt{(x-ae)^2 + y^2}$. We can substitute (6) and (7) into this:

$$\begin{aligned} r &= \sqrt{(a \cos E - ae)^2 + (a\sqrt{1-e^2} \sin E)^2} \\ &= \sqrt{a^2(\cos^2 E - 2e \cos E + e^2) + a^2(1-e^2) \sin^2 E} \\ &= a\sqrt{\cos^2 E - 2e \cos E + e^2 + \sin^2 E - e^2 \sin^2 E} \\ &= a\sqrt{(\sin^2 E + \cos^2 E) - 2e \cos E + e^2 - e^2 \sin^2 E} \\ &= a\sqrt{1 - 2e \cos E + e^2 - e^2(1 - \cos^2 E)} \\ &= a\sqrt{1 - 2e \cos E + e^2 \cos^2 E} \\ &= a\sqrt{(1 - e \cos E)^2} \\ \implies r &= a(1 - e \cos E). \end{aligned} \quad (9)$$

We now have a nice expression for r , which is great! We then substitute (6) and (9) into (8), with the goal of finding a relation between f and E :

$$\begin{aligned}\cos f &= \frac{x - ae}{r} \\ &= \frac{a \cos E - ae}{a(1 - e \cos E)} \\ \implies \cos f &= \frac{\cos E - e}{1 - e \cos E}.\end{aligned}\tag{10}$$

With the trig identity $\tan^2(\frac{f}{2}) = \frac{1 - \cos f}{1 + \cos f}$, we can substitute in (10):

$$\begin{aligned}\tan^2(\frac{f}{2}) &= \frac{1 - \frac{\cos E - e}{1 - e \cos E}}{1 + \frac{\cos E - e}{1 - e \cos E}} \\ &= \frac{\frac{(1 - e \cos E - \cos E + e)}{1 - e \cos E}}{\frac{(1 - e \cos E + \cos E - e)}{1 - e \cos E}} \\ &= \frac{1 - e \cos E - \cos E + e}{1 - e \cos E + \cos E - e} \\ &= \frac{(1 + e)(1 - \cos E)}{(1 - e)(1 + \cos E)}.\end{aligned}\tag{11}$$

Let's use the same trig identity again!

$$\begin{aligned}\frac{(1 + e)(1 - \cos E)}{(1 - e)(1 + \cos E)} &= \frac{1 + e}{1 - e} \tan^2(\frac{E}{2}) \\ \implies \tan(\frac{f}{2}) &= \sqrt{\frac{1 + e}{1 - e}} \tan(\frac{E}{2}).\end{aligned}\tag{12}$$

This tangent-based form is preferable to others because it effectively combines two equations (one for sine, one for cosine) into one since $\tan(\frac{f}{2}) = \frac{\sin(\frac{f}{2})}{\cos(\frac{f}{2})}$. Hassling with one equation instead of two makes our math simpler! Also, if we wanted to take the inverse tangent of both sides to find a direct expression for f , an added bonus is that the inverse tangent has nicer properties for calculating an angle in the correct quadrant (think `atan2` function in code). Regardless, we now have a clear path to finding an expression for $\frac{df}{dE}$! Let's take the derivative w.r.t. E of both sides of (12):

$$\begin{aligned}\frac{1}{2} \sec^2(\frac{f}{2}) \frac{df}{dE} &= \frac{1}{2} \sqrt{\frac{1 + e}{1 - e}} \sec^2(\frac{E}{2}) \\ \implies \frac{df}{dE} &= \sqrt{\frac{1 + e}{1 - e}} \frac{\sec^2(\frac{E}{2})}{\sec^2(\frac{f}{2})} \\ &= \sqrt{\frac{1 + e}{1 - e}} \frac{\sec^2(\frac{E}{2})}{1 + \tan^2(\frac{f}{2})}.\end{aligned}\tag{13}$$

Substituting (12) into the denominator to get rid of that pesky f on the right-hand side:

$$\begin{aligned}
 \sqrt{\frac{1+e}{1-e}} \frac{\sec^2(\frac{E}{2})}{1 + \tan^2(\frac{f}{2})} &= \sqrt{\frac{1+e}{1-e}} \frac{\sec^2(\frac{E}{2})}{1 + \frac{1+e}{1-e} \tan^2(\frac{E}{2})} \\
 &= \sqrt{\frac{1+e}{1-e}} \frac{1}{\cos^2(\frac{E}{2}) + \frac{1+e}{1-e} \sin^2(\frac{E}{2})} \\
 &= \sqrt{\frac{1+e}{1-e}} \frac{1-e}{(1-e)\cos^2(\frac{E}{2}) + (1+e)\sin^2(\frac{E}{2})} \\
 &= \frac{\sqrt{(1+e)(1-e)}}{\cos^2(\frac{E}{2}) + \sin^2(\frac{E}{2}) - e\cos^2(\frac{E}{2}) + e\sin^2(\frac{E}{2})} \\
 &= \frac{\sqrt{1-e^2}}{1 + e(\sin^2(\frac{E}{2}) - \cos^2(\frac{E}{2}))} \\
 \implies \frac{df}{dE} &= \frac{\sqrt{1-e^2}}{1 - e \cos E}. \text{ (double-angle trig identity)} \tag{14}
 \end{aligned}$$

Let's substitute this glorious equality, as well as (9) (our expression for r) and $b = a\sqrt{1-e^2}$, back into (5):

$$\begin{aligned}
 dM &= \frac{r^2}{ab} \frac{df}{dE} dE \\
 &= \frac{a^2(1 - e \cos E)^2}{a^2\sqrt{1-e^2}} \frac{\sqrt{1-e^2}}{1 - e \cos E} dE \\
 \implies dM &= (1 - e \cos E) dE. \tag{15}
 \end{aligned}$$

Let's take the integral of both sides:

$$\begin{aligned}
 \int dM &= \int (1 - e \cos E) dE \\
 \implies M &= E - e \sin E. \tag{16}
 \end{aligned}$$

Huzzah!

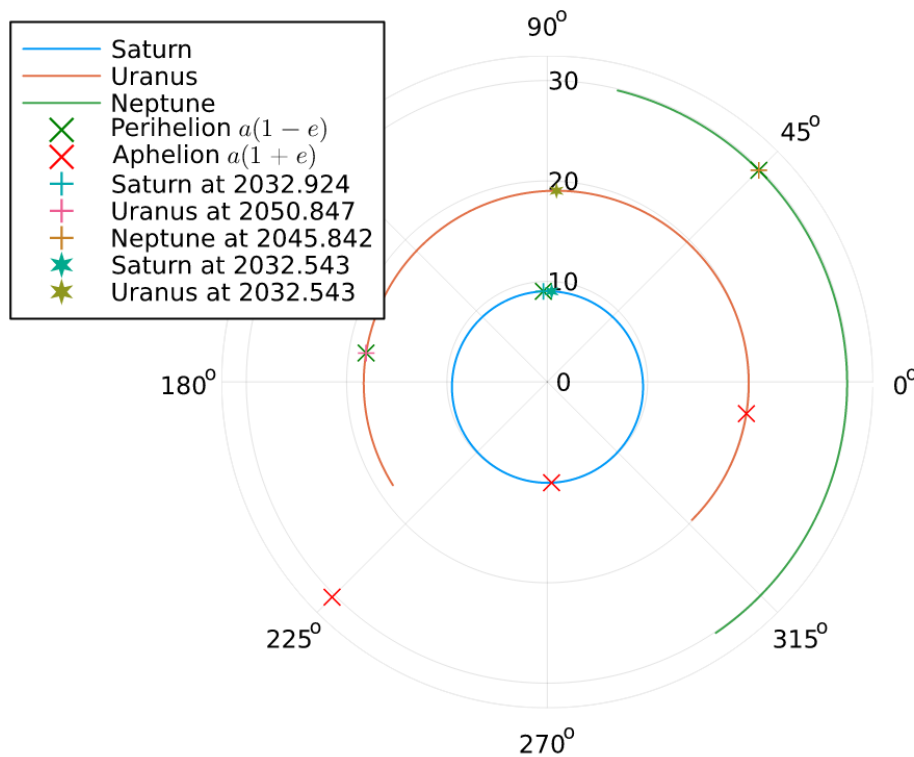
3 Build Keplerian Orbit Simulator

My simulator code is viewable at:

<https://github.com/nlizmi/astro4434/blob/main/ps1/sim.jl>

Results are shown below. First, the orbits of the three outer planets from J2000 to 2060. Classically-calculated (i.e., non-simulated) perihelion and aphelion are shown as x symbols, and the simulated perihelion passage times are shown with crosses (they line up very well!). Of note is that Saturn underwent more than one perihelion passage in this time period, but my algorithm happened to pick the one at 2032.924. The Saturn–Uranus conjunction time (2032.543) is shown with six-pointed stars.

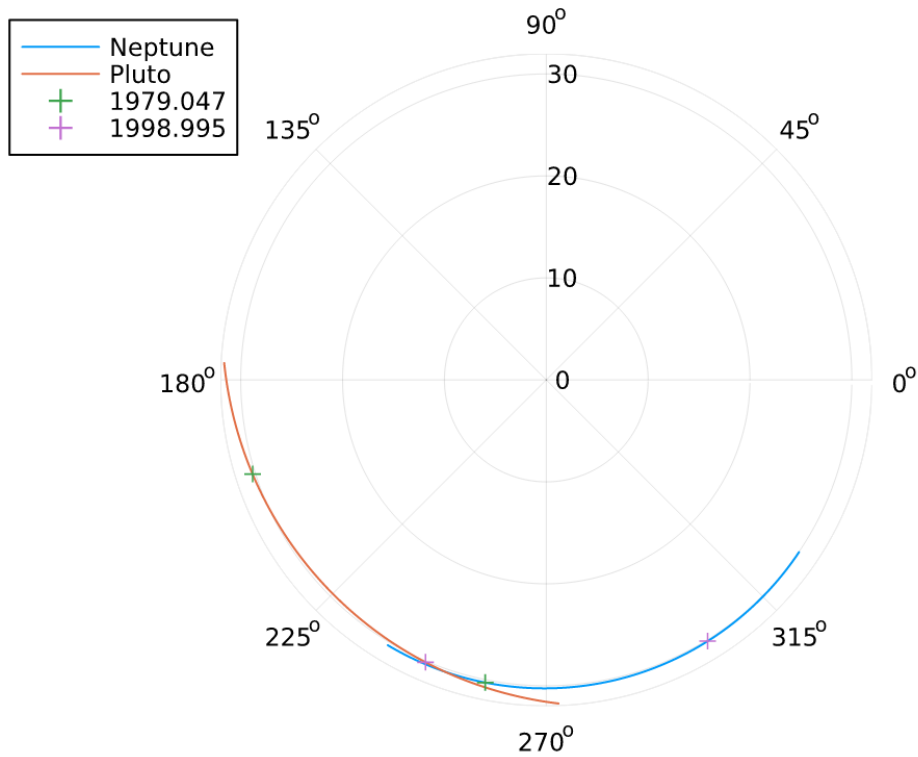
Orbits from 2000.0 to 2060.0 (r in AU)



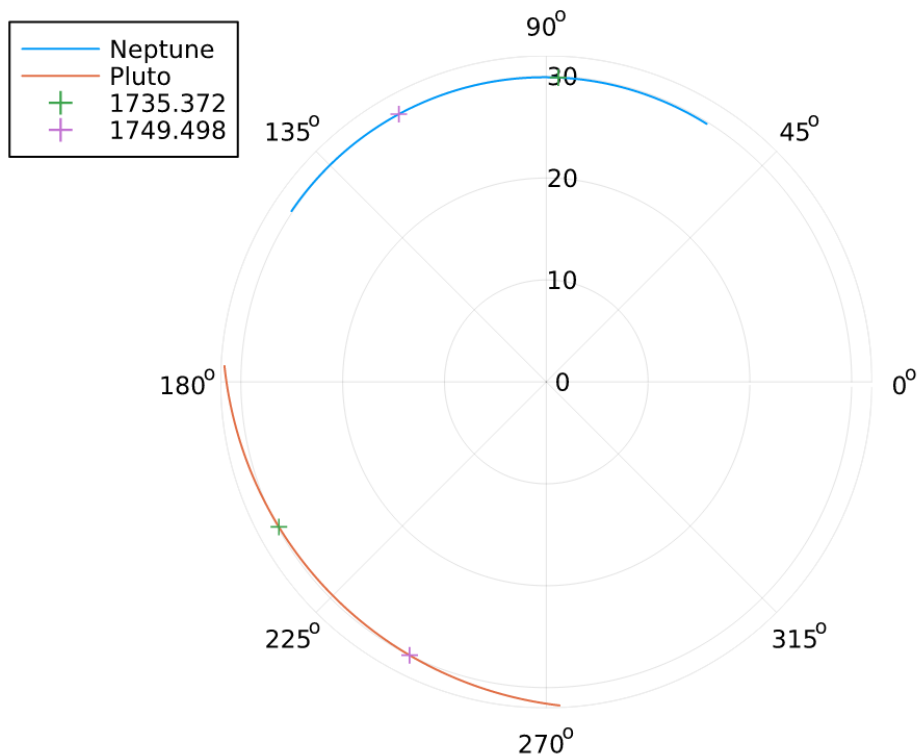
4 Pluto's Peculiar Orbit

Next are the plots of Neptune and Pluto. It may be difficult to see, but from the green crosses to the purple crosses in each plot (with dates indicated in the legends), Pluto's radial distance from the Sun is smaller than Neptune's. The last time this occurred was roughly one Pluto-period (~ 248 years) before, and it lasted for a shorter stretch of time because Neptune was at a point in its orbit where it was closer to the Sun. Neither Neptune nor Pluto had been discovered at this point in time (mid-1700s), so this phenomenon was not observed by 18th-century astronomers.

Orbits from 1970.0 to 2010.0 (r in AU)



Orbits from 1722.0 to 1762.0 (r in AU)



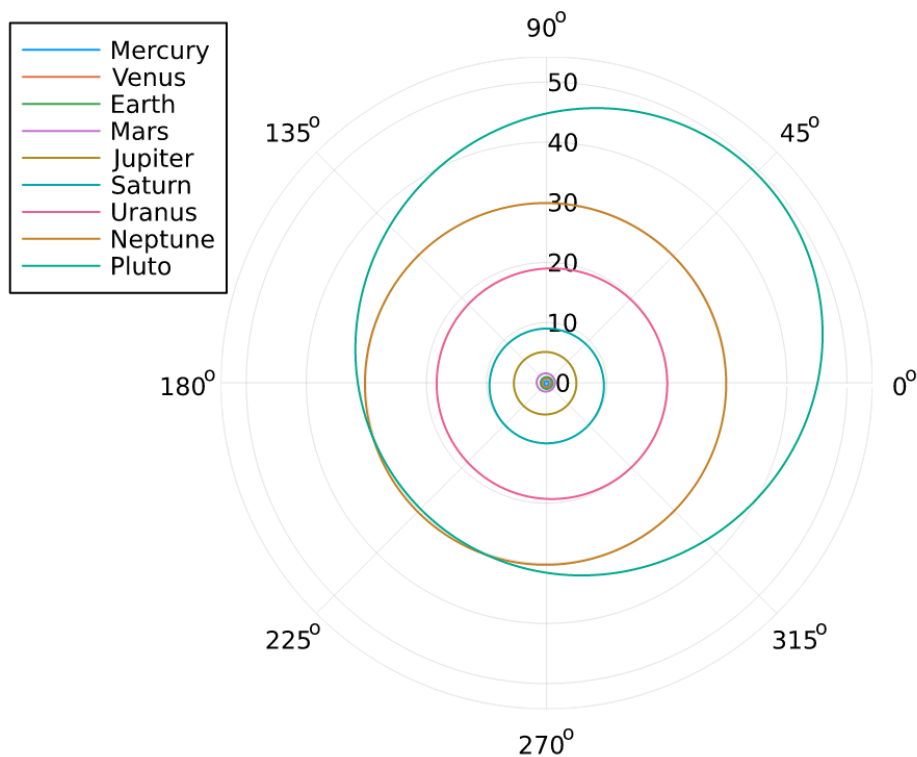
The discovery of Pluto and Neptune are interesting because they were predicted to exist

mathematically before they were directly observed. Uranus's motions were seemingly being perturbed in a way that the known planets could not account for according to Newton's laws, so the French astronomer Urbain Le Verrier (and separately the British astronomer John Couch Adams, about whom there is much controversy because many people think he made the discovery first) predicted the existence of an eighth planet as well as its position, mass, and orbit. Following Le Verrier's calculations, within a few months, Neptune was spotted by German astronomers within one degree of its predicted position in late September of 1846.

Despite this discovery, Uranus's orbital perturbations were still not fully reconciled, so the existence of yet another planet was theorized. A few individuals tried to spot it throughout the early 20th century, but it was not seen (or perhaps not recognized) until early 1930, when a spry youngster named Clyde Tombaugh identified it at Lowell Observatory using a machine called a blink comparator. Nice job, Clyde!

For fun, below I include a plot of the orbits of all eight planets plus Pluto. It's quite easy to see the spot where Pluto passes within Neptune's orbit here!

Orbits from 2000.0 to 2250.0 (r in AU)



5 GR Orbital Precession

For a circular orbit of Mercury, we can use:

$$v = \sqrt{\frac{GM_{\odot}}{a_{\text{Mer}}}}$$

given the semi-major axis of Mercury's orbit, $a_{\text{Mer}} = 5.791 \times 10^{10} \text{ m}$. Then:

$$\begin{aligned} \frac{v^2}{c^2} &= \frac{GM_{\odot}}{c^2 a_{\text{Mer}}} \\ &= \frac{(6.6743 \times 10^{-11} \text{ m}^3/\text{kg}/\text{s}^2)(1.989 \times 10^{30} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2 (5.791 \times 10^{10} \text{ m})} \\ &\approx 2.55 \times 10^{-8}. \end{aligned} \tag{17}$$

We now want Mercury's instantaneous velocity at radial distance $r = r_p = a(1 - e)$ in an eccentric orbit, for which we must use the Vis Viva equation. Given eccentricity $e_{\text{Mer}} = 0.2056$:

$$\begin{aligned} v^2 &= GM_{\odot} \left(\frac{2}{r} - \frac{1}{a_{\text{Mer}}} \right) \\ &= GM_{\odot} \left(\frac{2}{a_{\text{Mer}}(1 - e_{\text{Mer}})} - \frac{1}{a_{\text{Mer}}} \right) \\ \implies \frac{v^2}{c^2} &= \frac{GM_{\odot}}{c^2} \left(\frac{2}{a_{\text{Mer}}(1 - e_{\text{Mer}})} - \frac{1}{a_{\text{Mer}}} \right) \\ &= \frac{(6.6743 \times 10^{-11} \text{ m}^3/\text{kg}/\text{s}^2)(1.989 \times 10^{30} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2} \left(\frac{2}{(5.791 \times 10^{10} \text{ m})(1 - 0.2056)} - \frac{1}{5.791 \times 10^{10} \text{ m}} \right) \\ &\approx 3.871 \times 10^{-8}. \end{aligned} \tag{18}$$

Now, for Earth, we again use circular-orbit velocity and $a_{\text{Earth}} = 1.496 \times 10^{11} \text{ m}$:

$$\begin{aligned} \frac{v^2}{c^2} &= \frac{GM_{\odot}}{c^2 a_{\text{Earth}}} \\ &= \frac{(6.6743 \times 10^{-11} \text{ m}^3/\text{kg}/\text{s}^2)(1.989 \times 10^{30} \text{ kg})}{(2.998 \times 10^8 \text{ m/s})^2 (1.496 \times 10^{11} \text{ m})} \\ &\approx 9.873 \times 10^{-9}. \end{aligned} \tag{19}$$

We see that the ratio for Earth is about three times smaller than each of the ratios for Mercury, which makes sense since Earth orbits at about three times the distance that Mercury does. We therefore expect general relativity to have a greater impact on Mercury's orbit, though in general these ratios seem quite small. Let's see what the first-order GR

correction looks like for each planet!

$$\begin{aligned}
 \dot{\omega}_{\text{Mer}} &= \frac{3(GM_{\odot})^{\frac{3}{2}}}{a_{\text{Mer}}^{\frac{5}{2}}(1 - e_{\text{Mer}}^2)c^2} \\
 &= \frac{3((6.6743 \times 10^{-11} \text{ m}^3/\text{kg/s}^2)(1.989 \times 10^{30} \text{ kg}))^{\frac{3}{2}}}{(5.791 \times 10^{10} \text{ m})^{\frac{5}{2}}(1 - 0.2056^2)(2.998 \times 10^8 \text{ m/s})} \\
 &\approx 6.6053 \times 10^{-14} \text{ rad/s} \cdot \frac{206\,265''}{1 \text{ rad}} \cdot \frac{3.157\,76 \times 10^7 \text{ s}}{1 \text{ yr}} \\
 &\approx 0.4302''/\text{yr}.
 \end{aligned} \tag{20}$$

$$\begin{aligned}
 \dot{\omega}_{\oplus} &= \frac{3(GM_{\odot})^{\frac{3}{2}}}{a_{\oplus}^{\frac{5}{2}}(1 - e_{\oplus}^2)c^2} \\
 &= \frac{3((6.6743 \times 10^{-11} \text{ m}^3/\text{kg/s}^2)(1.989 \times 10^{30} \text{ kg}))^{\frac{3}{2}}}{(1.496 \times 10^{11} \text{ m})^{\frac{5}{2}}(1 - 0.0167^2)(2.998 \times 10^8 \text{ m/s})} \\
 &\approx \text{rad/s} \cdot \frac{206\,265''}{1 \text{ rad}} \cdot \frac{3.157\,76 \times 10^7 \text{ s}}{1 \text{ yr}} \\
 &\approx 0.0384''/\text{yr}.
 \end{aligned} \tag{21}$$

The answer for Mercury’s correction, shown in (20), is quite close to the observed difference of $0.425''/\text{yr}$. The small error may be due to several factors. For one, we are using many approximations—the GR correction itself is a first-order “weak-field” correction, meaning it must be an approximation of some more accurate GR correction for a stronger gravitational field, and we also make the assumption that M_{Mer} is small enough so as to be negligible compared to $M_{\odot}$, which is sort of true but may introduce some small inaccuracies in calculation. Perhaps if we had some more precise but more mathematically unwieldy equations to substitute values into, we would get the correct value for the GR correction. It’s also possible that GR is incorrect and doesn’t fully account for the physics going on here, but that’s less likely.

6 Tidal Heating of Io

We’re plugging values into this equation:

$$\frac{dE}{dt} = \frac{36\pi\rho^2 n^5 R_s^7 e^2}{19\mu Q}. \tag{22}$$

Below are values from the provided paper and from Wikipedia of all the useful parameters:

Parameter	Io	Enceladus
Density ρ	3.5 g/cm^3	1.61 g/cm^3
Mean motion n	$4.111 \times 10^{-5} \text{ rad/s}$	$5.307 \times 10^{-5} \text{ rad/s}$
Radius R_s	$1.8216 \times 10^8 \text{ cm}$	$2.521 \times 10^7 \text{ cm}$
Eccentricity e	0.0043	0.0047
Rigidity μ	$6.5 \times 10^{11} \text{ dyne/cm}^2$	$6.5 \times 10^{11} \text{ dyne/cm}^2$

I'm making the assumption that Enceladus's rigidity μ is the same as Io's as provided in the paper, which may or may not be accurate. Regardless, when we substitute these values for each planet into (22), we get the following numerical answers:

$$\left. \frac{dE}{dt} \right|_{Io} \approx \frac{1.621 \times 10^{21} \text{ erg/s}}{Q_{Io}} = \frac{1.621 \times 10^{14} \text{ W}}{Q_{Io}}. \quad (23)$$

$$\left. \frac{dE}{dt} \right|_{Enceladus} \approx \frac{1.429 \times 10^{15} \text{ erg/s}}{Q_{Enceladus}} = \frac{1.429 \times 10^8 \text{ W}}{Q_{Enceladus}}. \quad (24)$$

Assuming that $Q_{Io} = Q_{Enceladus}$, we therefore see that Io's tidal dissipation is roughly $10^6 \times$ greater than that of Enceladus.

We did not know for certain that Io was volcanically active before *Voyager I*'s observations; this paper by Peale, Cassen, and Reynolds was the first time this idea was really concretely explored. On that note, *Voyager I* taught us these three things about the Jupiter system, at the very least:

1. Jupiter has a faint ring system (only Saturn was known to have one at the time)
2. Jupiter's atmosphere and magnetosphere is more active than previously thought (Io plasma torus, lightning, etc.)
3. What the surfaces of the Galilean moons (and Amalthea) look like from up close, including Io's volcanoes and the interesting icy structure of Europa's surface.

From *Voyager II*, we learned a plethora of fascinating stuff about Enceladus. We found that while it has a largely cratered icy surface, its southern pole is almost entirely uncratered, suggesting the presence of a resurfacing mechanism and geological/hydrothermal activity. Furthermore, the southern pole has a series of long crevasses known as the Tiger Stripes that are hot relative to their surroundings and emit a series of icy plumes into space! This suggests the presence of a subsurface ocean with hydrothermal vents, which is super exciting from an astrobiological point of view and would be the first known extraterrestrial instance of hydrothermal vents.

Based on our calculations in (23) and (24), the tidal dissipation that creates Io's volcanic activity is much more powerful than that affecting Enceladus, but it is likely that tidal heating is the reason that Enceladus has a liquid subsurface ocean rather than it being a ball of ice, and it might be reasonable to assume that this tidal heating is also the source of the hydrothermal vents.